Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How load does it take to download the file?

Reasoning + Answer:

<think> 40% of the way through the download is at 80GB (200*40/100= 80) when she runs into the crash. 20 minutes is 20/60= 1/3 minute per GB. 1/3*80= 26.6GB per minute. To download the 80GB she missed she takes 80/26.6= 3 minutes. Now she has to redownload the entire 200GB file so her new total is 4*200= 800 minutes. She lost 20+3= 23 minutes. In 800 minutes she would have downloaded 800*.05= 40GB per minute. That means she can download 2+40= 42GB per minute. The total she could download 42*800= 33,600GB per minute. In 800 minutes she will download 33,600/800 = 42 minutes. </think><answer>33600 </answer>